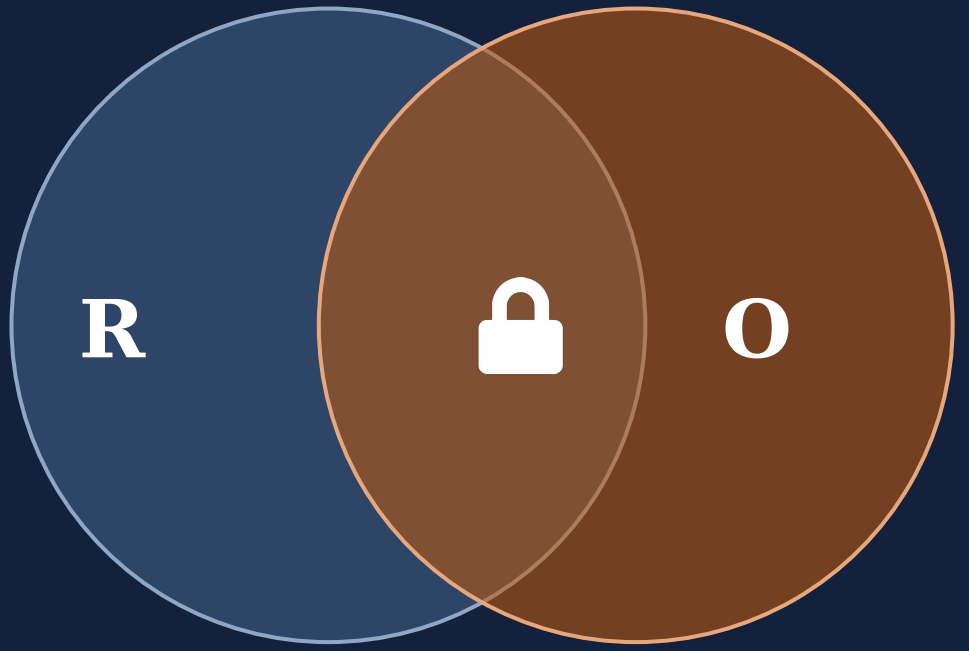
Privacy-Preserving Vertical Federated Linear Regression

When the organization holds the response: protocol, proofs and evaluation



Working draft · October 2026

1

The problem

Joint regression across data that cannot be pooled

Two datasets, one question, no pooling

Researcher R

- Smaller cohort with rich domain variables: health, education, surveys
- Wants to know whether these predict an administrative outcome
- Holds features X_R and identifiers

Organization O (e.g. StatCan)

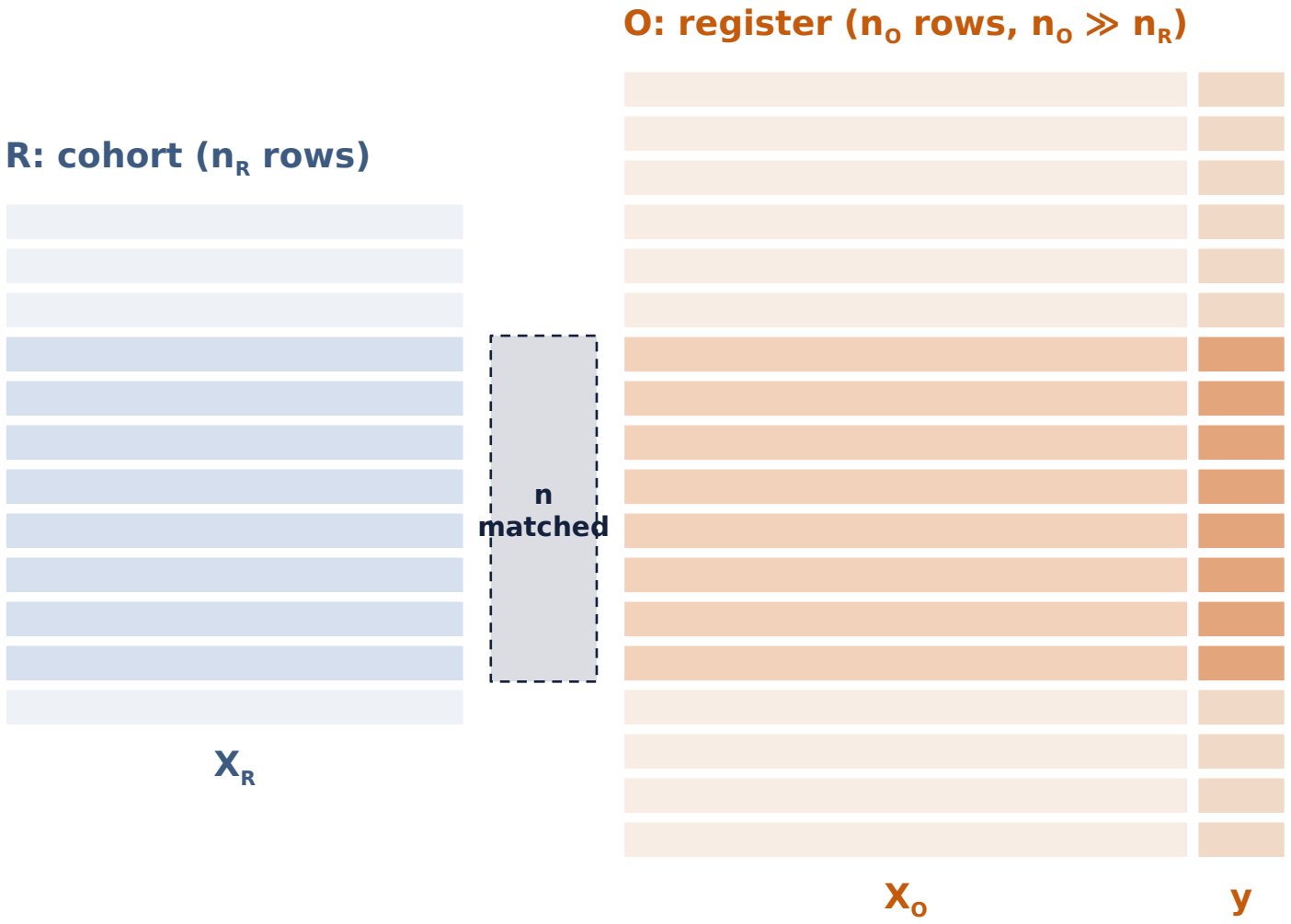
- Population register: tax filings, income, benefits
- Holds the outcome y and features X_O
- Cannot release records; law and governance bar direct sharing

6–8 weeks

of vetting and approvals today before a researcher can link a cohort to agency records in a Research Data Centre. Neither dataset alone answers the question.

Same people, different variables

Vertically partitioned data: overlapping rows, disjoint columns



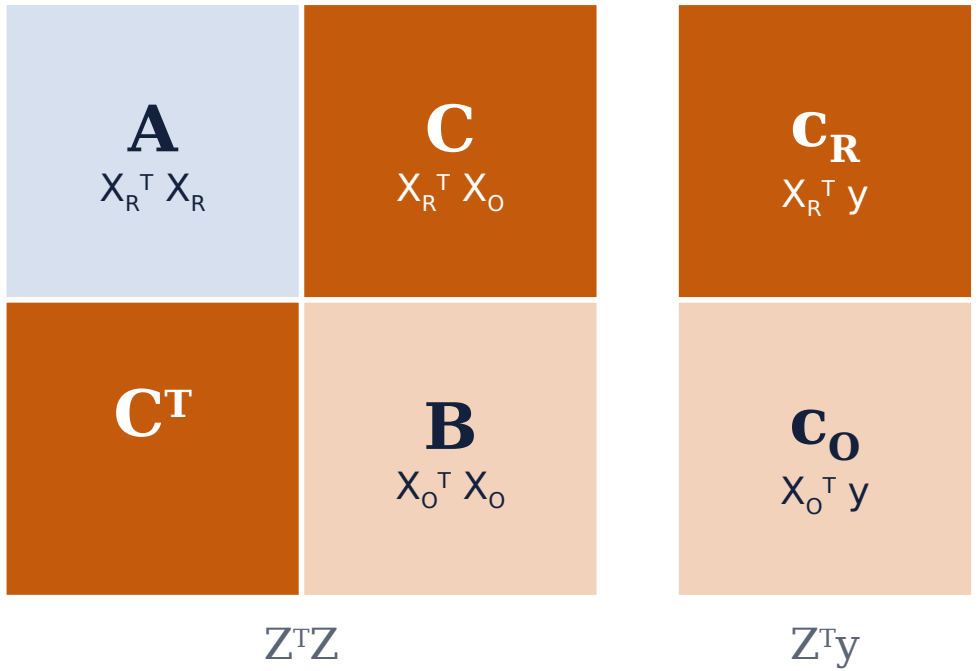
Goal

- Fit y on [X_R | X_O] over the n matched individuals
- Neither party sees the other's raw data
- O must not learn which of its records are in R's cohort
- The released model must not expose individuals

What has to be computed

OLS on the matched join needs a Gram matrix that spans both parties

$$\hat{\beta} = (Z^\top Z)^{-1} Z^\top y, \qquad Z = \left[\, X_R^I \;\middle|\; X_O^I \,\right]$$



R alone: exact, never leaves R

O's data, but only on matched rows (needs b)

Cross-party: needs both parties' data

Only A is purely R's. Everything that touches O's data is computed under encryption and noised by O before R sees it.

Three guarantees, honest-but-curious parties

Input privacy

Each party's view can be simulated from its own input and the agreed output. Neither sees the other's raw data.

Linkage privacy

O learns nothing about which of its records match R's cohort (only public sizes).

Output privacy

What R receives is differentially private (SIM-CDP) for each matched person's O-side data (x_O, y), with R's features held fixed (replace-one).

Parties follow the protocol but analyse everything they receive. In v3.1, R and O each hold a share of the decryption key (in v2, R held all of it). R learns the match set by design; O controls the privacy budget.

Where this work sits

Approach	Where the data sit	Record linkage	Output privacy	Gap for our setting
Central DP regression (AdaSSP, Wang 2018)	One trusted curator holds the joined table	Not needed	DP	Assumes the pooled data exist
VFL with HE (e.g. Yang et al. 2019)	Split across parties	Assumed or separate PSI	Usually none	Model can leak individuals
Private set intersection (Chen–Laine–Rindal)	Split across parties	Yes	Not applicable	Linkage only, no model
This work	Split; never pooled	Labeled PSI, private to O	DP, budget controlled by O	Narrower guarantee: protects O's side; R learns membership

Same statistical engine as AdaSSP (perturb the sufficient statistics, then solve a ridge), delivered without pooling, and with R's own block kept exact.

2

What we did

Reviewed the draft and rebuilt it; had the rebuild refereed, then rebuilt and proved it



1. Review

Draft paper and code checked line by line

2. Run

End-to-end over sockets: plaintext and OpenFHE

3. Attack

Three privacy reviews and a critique

4. Redesign

First revision (v2) and rewritten paper

5. Simulate

Accuracy and bias-correction studies

6. Benchmark

AdaSSP on 29 of Wang's UCI datasets

7. Referee

Journal-style review; key claims re-checked

8. Prove

Second revision (v3.1) with a proofs appendix

Critical finding: the noise did not protect the ciphertext

100%

of trials in which R identified the true dataset from O's reply: matched and unmatched records alike (real OpenFHE, 20/20).

73%

the most that ε = 1 differential privacy allows.

Why

- The draft added the noise as a plaintext scalar, which changes only c_0
- c_1 is a deterministic function of R's own ciphertexts, keys and O's data
- R, who holds the key, recomputes c_1 for each candidate dataset and compares

Fix

- Add a fresh encryption of the noise, which re-randomizes c_1
- Switch to the last modulus level, then flood the error term
- BFV linkage replies: statistical flooding; CKKS: DP-calibrated flooding
- **v3.1 goes further: the key is split, and O's decryption share carries the noise**

Review scorecard

Findings on the draft, how the first revision (v2) handled them, and where they stand in v3.1

Finding on the draft	Severity	First revision (v2)	Now (v3.1)
Plaintext-scalar noise leaks through the ciphertext	**Critical**	*Fresh encryption plus flooding*	Split key; O's decryption share carries the noise (proven)
Calibration step shares raw-data statistics	**Critical**	Per-party commitments, public constants only	Unchanged
Add/remove sensitivity vs replace-one adjacency (true ε up to ~2× the claim)	**High**	Replace-one Δ, closed form	Δ proven exact; $y^{\top}y$ released only where it is free
No privacy ledger across runs	**High**	*zCDP ledger at O*	(ρ, δ) ledger per register; flooding costs only δ
PSI reveals O's row order and bin loads	**High**	Secret permutation, public padding	Also OPRF-based; R's cohort capped and padded
Adaptive λ cancels the bias correction	**Validity**	*λ set from public inputs*	λ from a stage-1 release, independent of stage 2 (proven)
CKKS slot leak; HE parameter security	**Checked**	Not an issue with OpenFHE	CKKS scale must rise to $\approx 2^{61.5}$ (proofs)
R learns which of its cohort are in the register	**By design**	Stated in Definition 4	Unchanged

Grey: fixes that the referee later found incomplete (Part 5).

Contributions

1 Corrected data flow

R's Gram block A is computed locally, is never sent and never noised. The sensitivity loses its B_R^4 term.

2 Protocol v3.1

Split decryption key with the noise in O's share, replace-one DP, (ρ, δ) ledger, OPRF-based PSI, two-stage ridge, valid intervals.

3 Proofs

Output privacy (SIM-CDP) with a worst-case error bound, the exact replace-one sensitivity, and interval coverage to first order.

4 Evaluation

End-to-end runs, simulation studies of accuracy and of the bias correction, and a benchmark against AdaSSP.

3

Methodology

The first revision (v2), phase by phase

Protocol v2 at a glance

0 **Setup and commitments**

Only public constants cross

1 **Private linkage**

BFV labeled PSI; R learns the match set

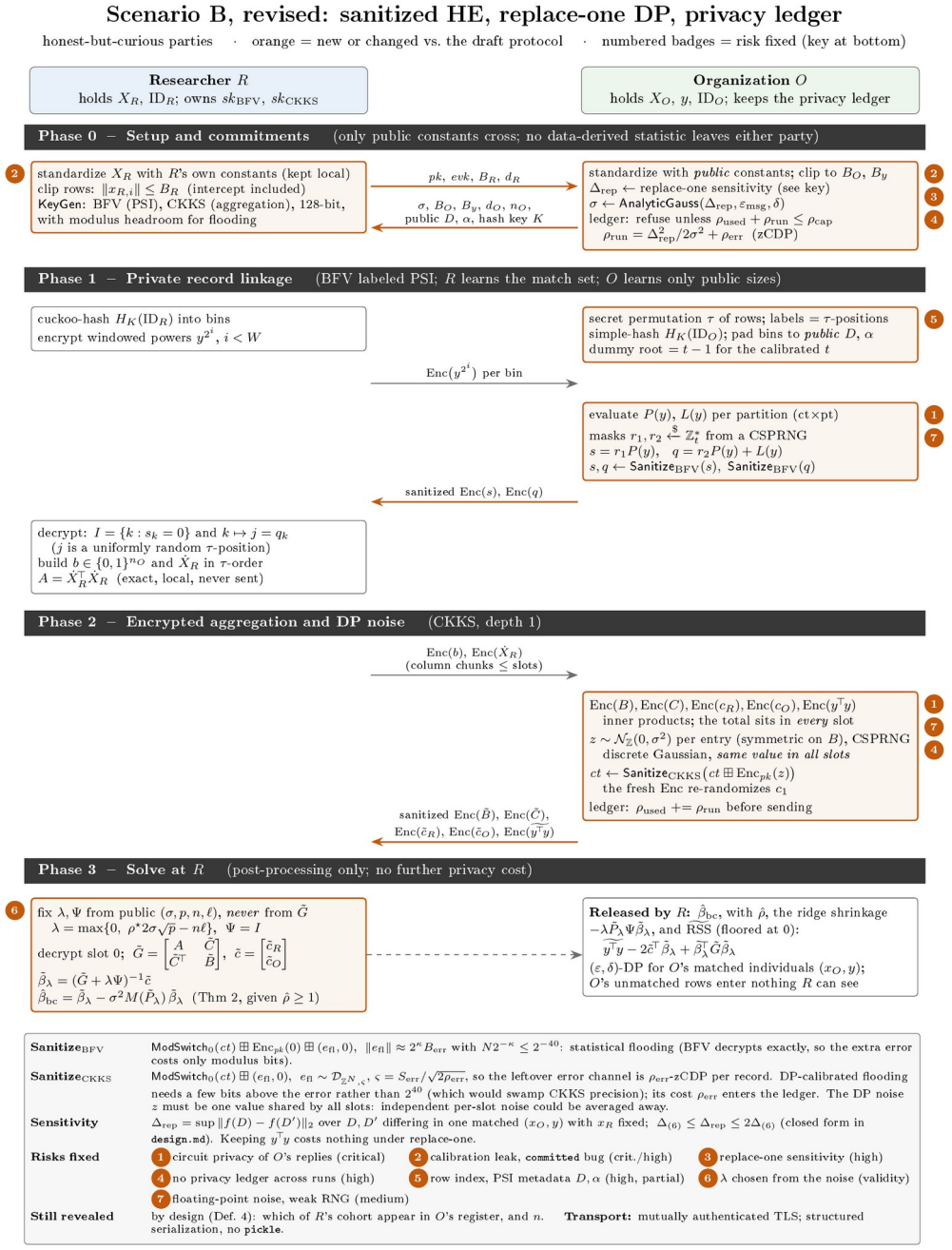
2 **Encrypted aggregation**

CKKS, depth 1; DP noise added by O

3 **Solve at R**

Fixed ridge, bias correction; post-processing

Orange in the figure marks what changed from the draft. Part 5 shows what v3.1 changes.



Phases 0–1: commitments and private linkage

0 Setup and commitments

- R: standardizes with its own constants (kept), clips rows to B_R, generates BFV and CKKS keys
- O: standardizes with public constants, clips to B_O and B_y
- O computes Δ_{rep} and σ, and refuses the run if the ledger would exceed its cap
- **Only public constants cross: no quantiles, clip rates or sample moments**

1 Labeled PSI with alignment

- O applies a secret permutation τ, so labels are random positions
- Keyed hashing; bins padded to a public degree D and count α
- R cuckoo-hashes its IDs and sends encrypted windowed powers
- O evaluates membership and label polynomials, masks, and sanitizes
- R learns I and the aligned positions; O learns only public sizes

Phase 2: encrypted aggregation

O multiplies R's encrypted vectors by its own plaintext columns (depth 1)

$$\mathrm{Enc}(B_{jk}) = \sum_i \mathrm{Enc}(b_i)\, X_{O,ij}\, X_{O,ik}$$

$$\mathrm{Enc}(C_{jk}) = \sum_i \mathrm{Enc}(\dot{X}_{R,ij})\, X_{O,ik}$$

$$\mathrm{Enc}(c_{O,k}) = \sum_i \mathrm{Enc}(b_i)\, X_{O,ik}\, y_i$$

$$\mathrm{Enc}(c_{R,j}) = \sum_i \mathrm{Enc}(\dot{X}_{R,ij})\, y_i$$

b: selection vector; $\dot{X}_R$: R's features in O's order (zero off the match set)

$A = \dot{X}_R^T \dot{X}_R$ stays with R

Exact, never transmitted, never noised.

One noise draw per entry

z is the same in every CKKS slot. Independent noise per slot could be averaged away.

Each released entry: $ct \leftarrow \mathrm{Sanitize}_{CKKS}(\, ct \boxplus \mathrm{Enc}_{pk}(z)\,)$, $z \sim$ discrete Gaussian $N_{\mathbb{Z}}(0, \sigma^2)$ from a CSPRNG

Calibrating the noise, and keeping count

Replace-one sensitivity

The adjacency replaces one matched person's (x_O, y). The draft used the add/remove bound, which understates Δ.

(B_R, B_O, B_y)	Δ_{add}	Δ_{rep}	Ratio
(1, 1, 1)	2.24	3.16	1.41
(3.16, 3, 3)	20.6	29.4	1.43
(√5, √5, 5)	30.4	31.6	1.04
Simulation population	58.2	86.2	1.48

$\Delta_{add} \le \Delta_{rep} \le 2\Delta_{add}$. Adding y^Ty costs nothing in these rows, but not always: it is free if $B_y^2 \le 2(B_R^2 + B_O^2)$. At (1, 1, 5), $\Delta = 26.1$ with y^Ty and 14.1 without.

σ from the analytic Gaussian

Smallest σ meeting (ε, δ) for the whole stacked release (Balle–Wang). Discrete Gaussian at the CKKS resolution.

Privacy ledger at O

O refuses a run once the spend would pass its cap; re-runs and inference queries count. v2 charged ρ_{err} once per run, though one record moves the error of all 63 replies. v3.1 tracks (ρ, δ): flooding costs only δ.

Sanitizing O's replies in v2 (circuit privacy)

R holds the key, so every reply must reveal nothing beyond its decrypted value

1 Fresh encryption

$ct \boxplus Enc_{pk}(z)$ makes c_1 computationally uniform: it no longer depends on O's data.

2 Modulus switch

Drop to the last level, which scales down the data-dependent error.

3 Flood the error

BFV: statistical flooding with $N \cdot 2^{-\kappa} \leq 2^{-40}$. CKKS: DP-calibrated flooding with width $\varsigma = S_{err} / \sqrt{(2\rho_{err})}$.

What v2 relied on

A flood sized as a DP mechanism, only a few bits above the error, to save CKKS precision; and the fresh encryption to hide c_1 from the key holder.

What the referee and the proofs found

- S_{err} is as large as the whole error, and its cost was charged once, not per reply
- Decrypting, R sees a hint on the fresh randomness: not a plain RLWE reduction
- v3.1 floods the full error inside O's decryption share. Proven, at a CKKS scale of $2^{61.5}$ ($2^{53.7}$ with circuit changes)

Phase 3: fixed ridge and zero-budget bias correction

Ridge, fixed before decryption

$$\lambda = \max\{\,0,\; 2\rho^{*}\sigma\sqrt{p} - n\ell\,\}, \qquad \Psi = I$$

Release gate

$$\hat{\rho} = \lambda_{\min}(\tilde{G} + \lambda I)\,/\,(2\sigma\sqrt{p}) \;\geq\; 1$$

Private ridge estimate

$$\tilde{\beta}_{\lambda} = (\tilde{G} + \lambda I)^{-1}\,\tilde{c}$$

Bias correction (optional in v3.1; off by default)

$$\hat{\beta}_{\mathrm{bc}} = \tilde{\beta}_{\lambda} - \sigma^{2} M(\tilde{P}_{\lambda})\,\tilde{\beta}_{\lambda}$$

Theorem (conditional bias)

$$\mathbb{E}[\,\tilde{\beta} \mid \mathcal{E}_{\eta}\,] - \hat{\beta} = \sigma^{2} M(P)\,\hat{\beta} + O(\sigma^{4}) + O\big(e^{-p(\eta\rho-1)^{2}/2}\big)$$

$$M(P) = \big(P^{2} + \mathrm{tr}(P)\,P - P\,\mathrm{diag}(P)\big)\,\Pi_{O} + \big(P\,\Pi_{O}P + \mathrm{tr}(\Pi_{O}P)\,P\big)\,\Pi_{R}$$

The plain mean does not exist, hence the conditioning on $\mathcal{E}_{\eta}$.

Why λ must not depend on the noise

Chosen from G̃, λ moves with the noise, and the correction removed none of the bias (1.0–1.6× left); with fixed λ it removes 87–98%. But ℓ needs the joint Gram, so in practice $\ell = 0$. v3.1 picks λ from a separate stage-1 release instead.

How we evaluate

Simulation population

Features on real-world scales (income, age, binary indicators, counts), correlated. Public standardization constants; committed 99.5% bounds; $R^2 \in \{0.3, 0.5, 0.9\}$.

Monte Carlo

n from 100 to 100k matched records; $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, $\delta = 10^{-5}$; 200 replicates, each with new data and a new DP release.

Metrics

RMSE against the true β (standardized); per-coefficient efficiency ratio; interval coverage; MSE ratio corrected/uncorrected.

Benchmark

29 UCI datasets from Wang's published run, with his preprocessing; AdaSSP as published and under our guarantee; 10-fold test MSE.

4

Results

Measured with the v2 release; what is still to run

The cryptographic layer is exact

Draft protocol, run end-to-end over sockets with plaintext and real OpenFHE backends

0

misaligned rows

PSI recovered exactly the true match set in all 6 cohorts, plaintext and OpenFHE

$\approx 10^{-10}$

max |β − plaintext answer|

noise off, against the plaintext ridge target at the same λ (10^{-3}): encryption adds no error

≈13 s

end-to-end on OpenFHE

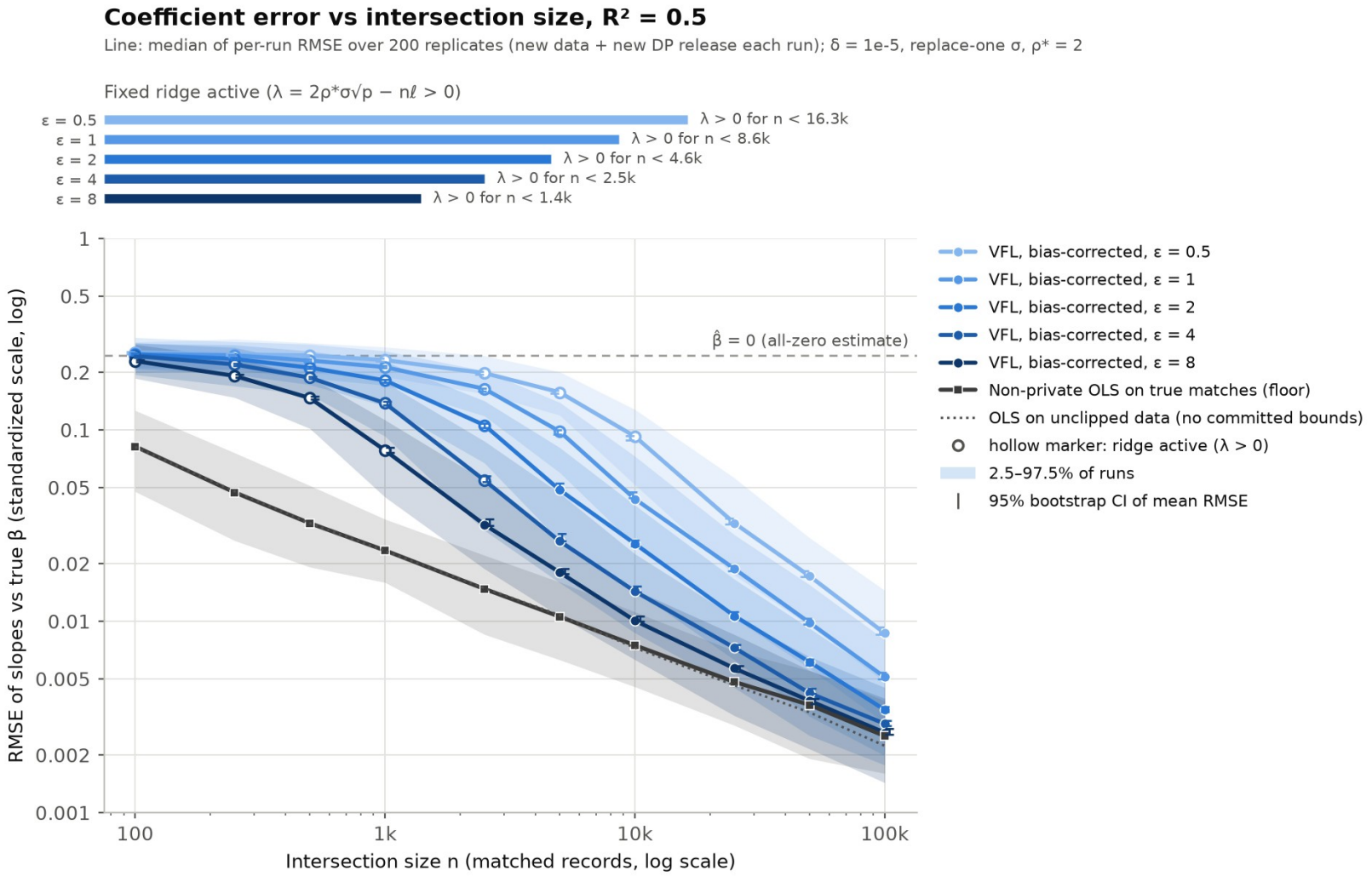
n = 400 matched out of n_O = 4,000 records (OpenFHE 1.5.1)

Cost of protocol v3.1 in OpenFHE

To be run

Register size n_o	d	PSI time (s)	Aggregation time (s)	Traffic (MB)	Release overhead
4,000	10	*tbd*	*tbd*	*tbd*	*tbd*
32,000	10	*tbd*	*tbd*	*tbd*	*tbd*
65,536	20	*tbd*	*tbd*	*tbd*	*tbd*
10^6 (chunked)	20	*tbd*	*tbd*	*tbd*	*tbd*

Accuracy against intersection size



Within 2× of non-private OLS at

ε = 8	**n ≈ 3.2k**
ε = 4	**n ≈ 9.4k**
ε = 2	**n ≈ 34k**
ε = 1	**n ≈ 105k**

- Small n: the ridge shrinks $\hat{\beta}$ towards 0
- Then the DP error falls like 1/n and merges into the OLS floor
- Run-to-run spread is about ×2.5–3

Key numbers

Median RMSE of the standardized slopes against the true β (ratio to the OLS floor), $R^2 = 0.5$

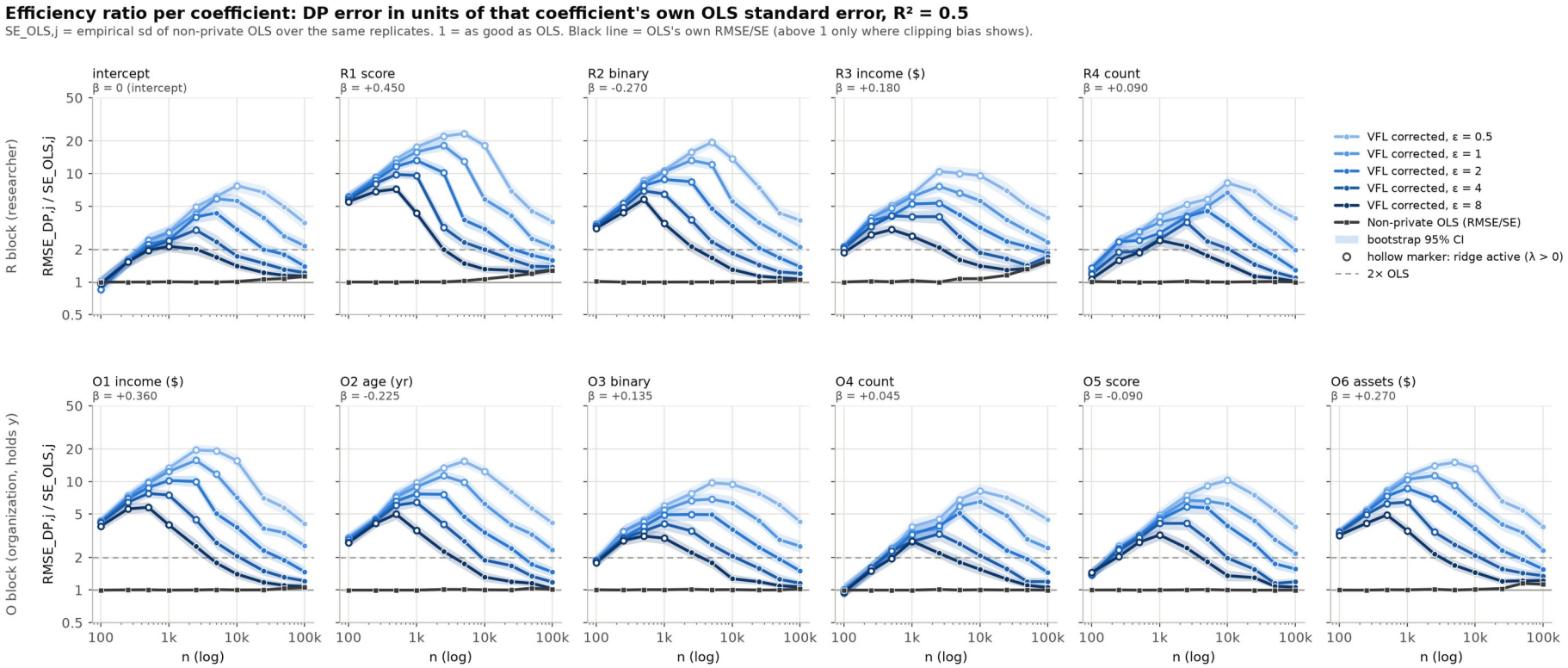
n	ε = 0.5	ε = 1	ε = 2	ε = 4	ε = 8	OLS floor
1,000	0.231 (9.9×)	0.213 (9.1×)	0.181 (7.7×)	0.138 (5.9×)	0.078 (3.3×)	0.023
10,000	0.092 (12×)	0.044 (5.8×)	0.025 (3.4×)	0.014 (1.9×)	0.010 (1.3×)	0.0075
100,000	0.0087 (3.4×)	0.0051 (2.0×)	0.0034 (1.4×)	0.0029 (1.2×)	0.0026 (1.0×)	0.0025

n at which VFL comes within 2× of the OLS floor

R^2	ε = 0.5	ε = 1	ε = 2	ε = 4	ε = 8
0.3	≈240k*	75k	21k	7.1k	1.9k
0.5	≈380k*	≈105k*	34k	9.4k	3.2k
0.9	>100k	>100k	66k	30k	13k

** extrapolated beyond n = 100k*

Comparing coefficients on different scales



Raw scale

Coefficients span 5.6 orders of magnitude (≈$0.2 per $ income vs ≈$10k per binary). A pooled raw RMSE is 87% driven by two binary coefficients.

Efficiency ratio

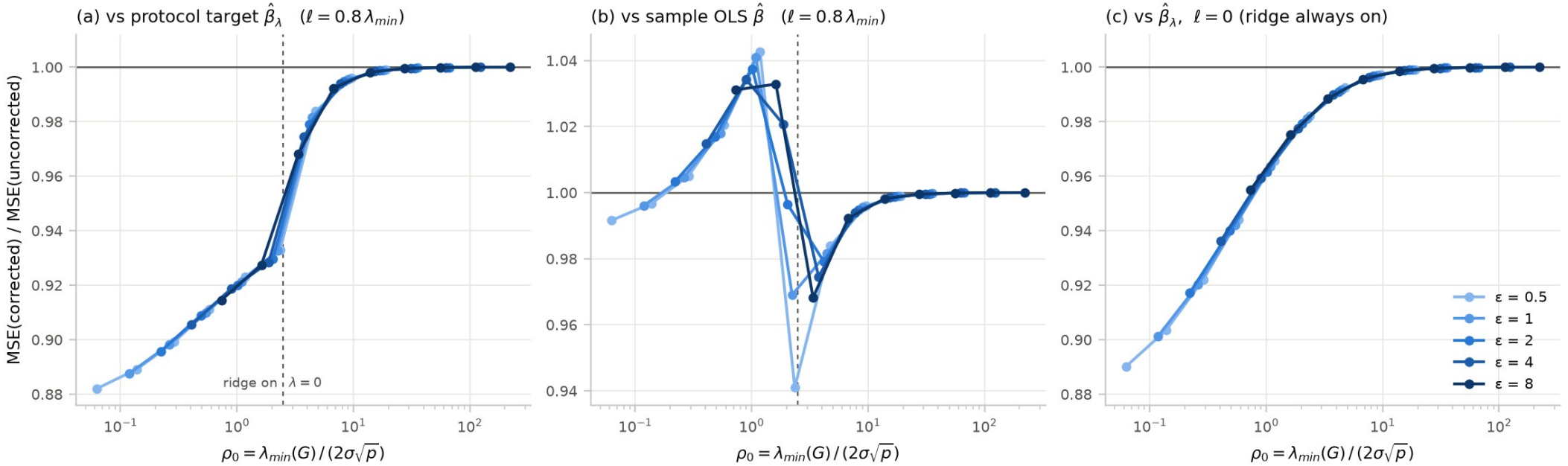
$RMSE_{DP,j}$ / $SE_{OLS,j}$: unit-free. A ratio r means the release is about as accurate as OLS on n/r^2 records.

Result

Once the ridge is off, DP costs about the same for every coefficient (±15%); O-side coefficients run ~10% higher.

Does the bias correction matter?

Barely: it acts mostly as extra shrinkage, so v3.1 makes it optional and off by default



−7 to −12%

MSE vs the protocol target when the ridge is on. The correction also shrinks noise; it never raised this MSE (90 settings).

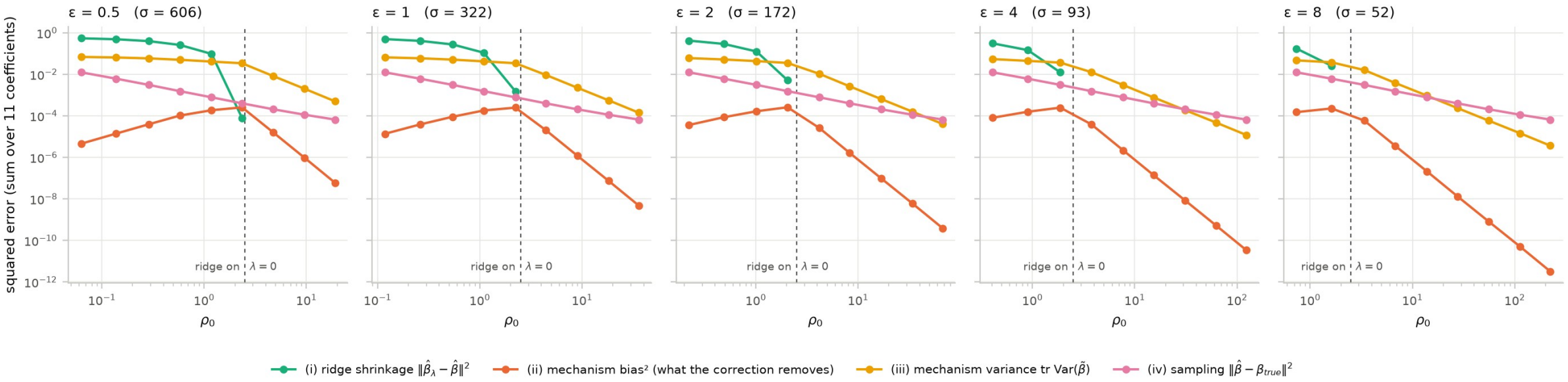
+0.3 to +4.3%

MSE vs OLS under a moderate ridge: the correction restores shrinkage that noise had offset.

≥ 130

averaged releases before the bias alone would matter; one release: at most 0.75% of MSE.

Where the error comes from when σ is large



Error anatomy

- Heavy ridge: 72–90% of the error is shrinkage, 11–31% noise variance
- Mechanism bias²: at most 0.75% anywhere
- At large n, clipping shifts the target (every DP method pays this)

Inference

- v2, λ = 0: 95% intervals cover 0.94–0.96
- v2, ridge on: coverage collapses (42–81% at $\ell = 0$)
- **v3.1: Wald intervals for the ridge target cover 0.944–0.962 (target 0.955) with the ridge on; the β intervals over-cover**

Benchmark against AdaSSP (Wang 2018)

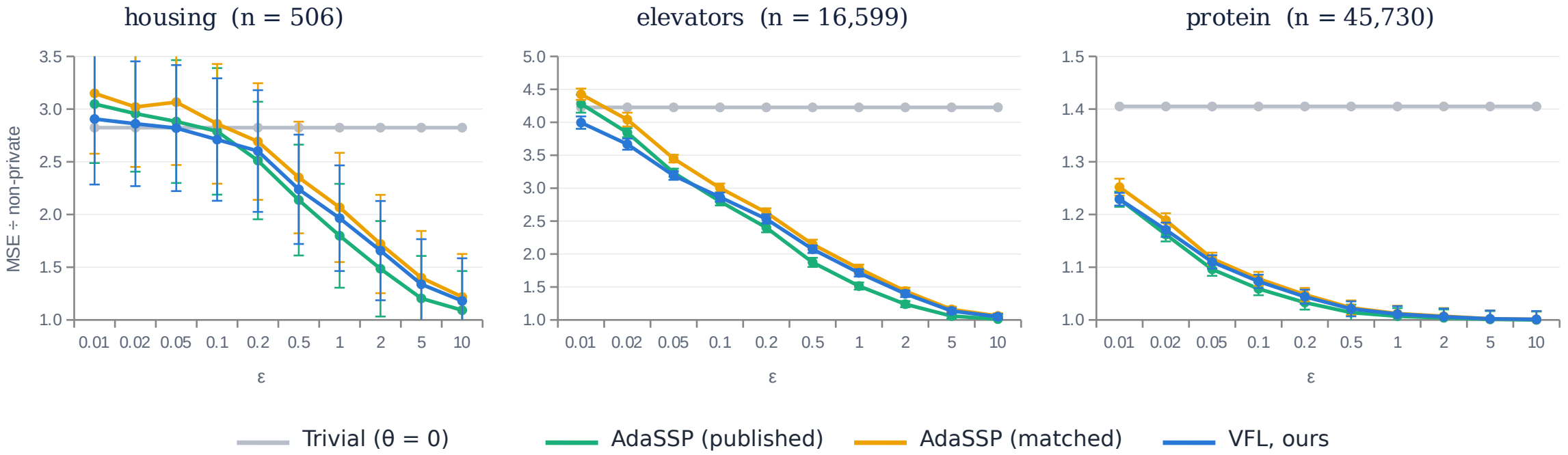
Test MSE at $\varepsilon = 1$, $\delta = 10^{-6}$, 10-fold CV. Ratio to non-private in brackets; bold = best private method

Dataset	n	d	Non-private	AdaSSP (published)	AdaSSP (matched)	VFL, ours	VFL, uncorrected
housing	506	13	0.0396	**0.0713 (1.80×)**	0.0819 (2.07×)	0.0779 (1.97×)	0.0773 (1.95×)
wine	1,599	11	0.0201	**0.0338 (1.68×)**	0.0391 (1.95×)	0.0370 (1.84×)	0.0368 (1.83×)
airfoil	1,503	5	0.0533	**0.0579 (1.09×)**	0.0625 (1.17×)	0.0617 (1.16×)	0.0616 (1.16×)
concrete	1,030	8	0.0444	**0.0657 (1.48×)**	0.0774 (1.74×)	0.0751 (1.69×)	0.0745 (1.68×)
bike	17,379	17	0.0279	**0.0288 (1.03×)**	0.0299 (1.07×)	0.0295 (1.06×)	0.0295 (1.06×)
elevators	16,599	18	0.00868	**0.0131 (1.51×)**	0.0154 (1.78×)	0.0149 (1.71×)	0.0147 (1.70×)
pol	15,000	26	0.135	**0.140 (1.04×)**	0.146 (1.08×)	0.144 (1.06×)	0.143 (1.06×)
kin40k	40,000	8	0.0632	**0.0633 (1.00×)**	0.0633 (1.00×)	0.0633 (1.00×)	0.0633 (1.00×)
protein	45,730	9	0.119	**0.120 (1.01×)**	0.120 (1.01×)	0.120 (1.01×)	0.120 (1.01×)

Published AdaSSP is more private than our release, not less: at nominal $\varepsilon = 1$ it is $\mu = 0.198$ Gaussian DP ($\varepsilon \approx 0.82$) under whole-record adjacency, against our $\mu = 0.237$ ($\varepsilon = 1.00$) under the weaker O-side adjacency. Its noise is lower because of Wang's composition and the joint row bound ($\Delta^2 = 4.5$, ours 10). Matched = AdaSSP's adaptive λ under our guarantee; idealized, since its λ_{min} step needs X^TX in the clear.

Across privacy levels

Test MSE ÷ non-private, three datasets of increasing size; bars: 95% CI of the mean over the 10 folds



Ours better / worse (of 29)	ε = 0.1	ε = 1	ε = 10
vs AdaSSP (published)	14 / 8	6 / 19	1 / 21
vs AdaSSP (matched)	27 / 0	27 / 0	21 / 3
Bias-corrected vs uncorrected	14 / 8	6 / 19	3 / 19

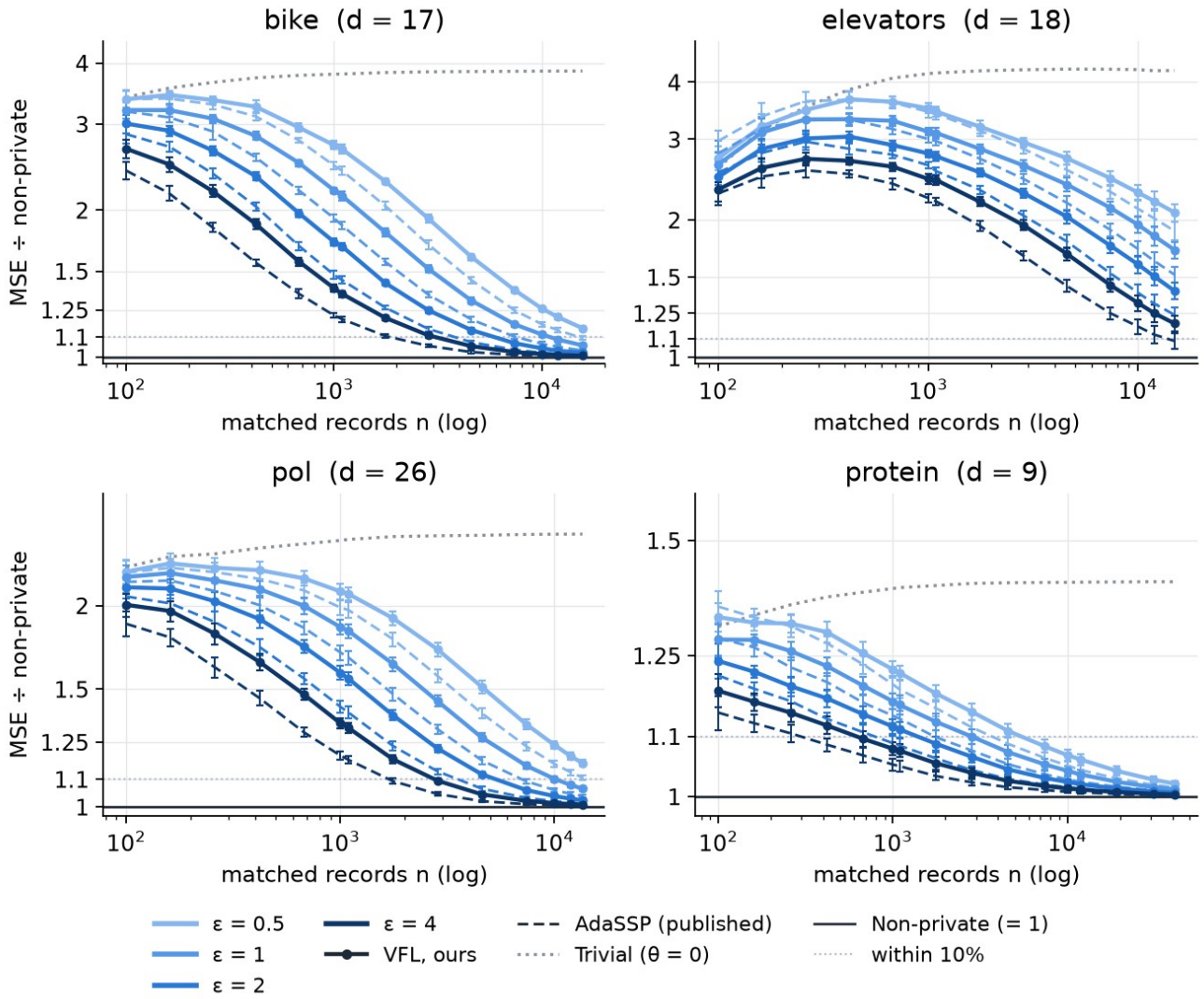
Paired over folds, 95% CI; the rest show no significant difference.

From ε ≈ 0.5 published AdaSSP wins (median 4% lower MSE at ε = 1); the gap shrinks as n grows. Read the wins with care:

- at ε = 0.1 both often lose to θ = 0 (ours on 13 of 29, AdaSSP on 15)
- the matched baseline uses only ¾ of its μ budget, so 27 / 0 overstates the fixed ridge

Intersection size on real data

Test MSE ÷ non-private at the same n; nested subsets of the training folds; bars: 95% CI over folds



Matched records to come within 10% of non-private

Ours (AdaSSP published)

	ε = 0.5	ε = 1	ε = 2	ε = 4
bike	**>15k** (15k)	**11k** (7.5k)	**5.9k** (3.8k)	**3.1k** (1.8k)
elevators	**>14k** (>14k)	**>14k** (>14k)	**>14k** (>14k)	**>14k** (14k)
pol	**>13k** (>13k)	**10k** (7.0k)	**5.3k** (3.5k)	**2.8k** (1.7k)
protein	**5.4k** (3.7k)	**2.8k** (1.7k)	**1.4k** (784)	**619** (295)

- Doubling ε roughly halves the records needed
- Ours needs about 1.5–2× the records AdaSSP needs for the same cost
- elevators stays above 10% at every size available (up to 14.9k)

Calibration in practice: from ε to σ and λ

Every input is public and committed before any data are seen

1 δ

$\delta \le 1/(10N)$

N = people O protects. $10^{-5} \to 10^{-8}$ raises σ by only 25–50%

2 Budget

one release: (ε, δ)

k releases:

$\rho = \left(\sqrt{L+\varepsilon} - \sqrt{L}\right)^2 / k$

L = ln(1/δ); zCDP adds up across releases

3 Δ

$\Delta^2 = \frac{1}{2}(2O + R + Y)^2 + 2RY$

R, O, Y = B_R^2, B_O^2, B_y^2 (usual case, exact). $y^T y$ is free if Y ≤ 2(R + O)

4 σ

$\sigma = \Delta \cdot s(\varepsilon, \delta)$

or $\Delta/\sqrt{2\rho}$ for k releases

s: analytic-Gaussian multiplier (table below)

5 λ

$\lambda = \max\{0, 2\rho^* \sigma \sqrt{p} - n\ell\}$

$\rho^* = 2.5$; ℓ public, else 0; ridge off from $n = 2\rho^*\sigma\sqrt{p} / \ell$. v3.1: λ from stage 1

σ/Δ δ \ ε	0.1	0.25	0.5	1	2	4	8
$\delta = 10^{-5}$	30.7	13.3	7.03	3.73	1.99	1.08	0.6
$\delta = 10^{-6}$	36.3	15.4	8.06	4.22	2.23	1.19	0.653
$\delta = 10^{-8}$	45.9	19.1	9.86	5.1	2.65	1.4	0.748

Planning formula

$$\sigma \approx \Delta \left(\sqrt{2\ln(1/\delta)} - 1\right) / \varepsilon^{0.92}$$

Within −8% to +3% of exact for ε ∈ [0.1, 8], δ ∈ $[10^{-8}, 10^{-5}]$. For planning only: O computes the exact σ for each release.

From σ to the matched records you need

Two scaling laws, exact at first order; one pilot figure carries over to any ε

Prediction

$$n \propto \sigma \propto \Delta/\varepsilon$$

With $\ell = 0$ the ridge is proportional to σ, so the excess test error depends on n only through n / σ. Doubling ε roughly halves the records.

Coefficients

$$n \propto \sigma^2 \propto (\Delta/\varepsilon)^2$$

Once the ridge is off, DP error / sampling error ≈ $\sigma^2(1+\|\beta\|^2)\ \mathrm{tr}(\Sigma^{-2}) / (n\, s^2\, \mathrm{tr}(\Sigma^{-1}))$. Doubling ε cuts the records by about 3.5×.

Target	Setting	First-order calculator	Transfer (× σ or × σ²)	Observed
Prediction within 10%	bike, ε = 2 (from ε = 1)	5.7k	5.8k	**5.9k**
Prediction within 10%	pol, ε = 4 (from ε = 1)	2.8k	2.9k	**2.8k**
Prediction within 10%	protein, ε = 4 (from ε = 1)	747	794	**619**
Coefficients within 2× OLS	R^2 = 0.5, ε = 4 (from ε = 2)	11k	10.0k	**9.4k**
Coefficients within 2× OLS	R^2 = 0.5, ε = 8 (from ε = 2)	3.3k	3.1k	**3.2k**
Coefficients within 2× OLS	R^2 = 0.3, ε = 1 (from ε = 2)	78k	74k	**75k**

Prediction: real-data sweep; coefficients: simulation. Breaks down for ill-conditioned designs (use the calculator's eigen-sums), at high R^2 where clipping bias dominates, and at small n. Flooding noise and the discrete Gaussian are not modelled.

What each choice costs

Defaults: $\delta = 10^{-6}$, $\rho^* = 2.5$, ℓ from public data, tight bounds, k fixed in advance

Change	σ	Records for prediction	Records for coefficients
Bounds 2× closer to the data (no extra clipping)	× 1/4	× 1/4	× 1/16
Double ε	× 0.52–0.55	× 0.52–0.55	× 0.27–0.30
δ from 10^{-6} to 10^{-8}	× 1.15–1.26	× 1.15–1.26	× 1.3–1.6
Same budget over k = 5 releases instead of 1	× 2.6–2.9	× 2.6–2.9	× 6.5–8.7
Split a unit row bound, $B_R^2 = B_O^2 = ½$ (if little extra clipping)	× 0.64	× 0.64	× 0.41

Worked example: the simulation population at n = 10,000, $\delta = 10^{-6}$

$B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $p = 11$, $\ell = 0.494$, $\rho^* = 2.5$

ε	Δ	σ, one release	zCDP cost ρ	λ at n = 10k	Ridge off from n	σ, each of 5 releases
0.5	86.2	694	0.0077	6577	23k	2044
1	86.2	364	0.0280	1099	12k	1031
2	86.2	192	0.1005	0	6.5k	524
4	86.2	103	0.3510	0	3.5k	270
8	86.2	56	1.1728	0	1.9k	143

5

Second revision: protocol v3.1

A referee review, a redesign for honest-but-curious parties, and proofs

Referee review: findings and answers

Recommendation: reject and resubmit. Every major comment accepted; the numerical claims re-checked

Major comment	Verdict	In v3.1
Privacy theorem unproven: S_{err} unbounded; a hint to the key holder; no computational DP notion; error cost charged once; finite precision undefined	**Accepted**	Split key; full error flooded inside O's share; SIM-CDP; (ρ, δ) ledger; lattice quantization. Proven
The semi-honest R is also the party DP protects against	**Accepted**	Stated: honest-but-curious only. Malicious R needs ZK input proofs (extension)
Revised protocol neither implemented nor costed	**Accepted**	**Open: OpenFHE implementation is next**
Privacy levels ignore the error channel	**Accepted**	End-to-end (ρ, δ) accounting, tight zCDP conversion
AdaSSP comparison mischaracterized	**Verified**	Published AdaSSP is more private (ε ≈ 0.82 vs 1.00); caveat replaced
No valid inference under $\ell = 0$ (coverage 42–81%)	**Verified**	Two-stage ridge; Wald interval for θ_λ, bias-aware interval for β
Bias theory correct but of modest value	**Partly**	Correction optional, off by default
y^Ty is not always free	**Verified**	Released only where free; exact criterion proven

Protocol v3.1 at a glance

Split decryption key

$s = s_R + s_O$. Neither party can decrypt alone; R uploads under s_R and O converts

Noise in O's decryption share

O sends $(c_1', h_O = c_0' + c_1' s_O + \eta)$ and never c_0'. η floods the whole ciphertext error

Two-stage ridge

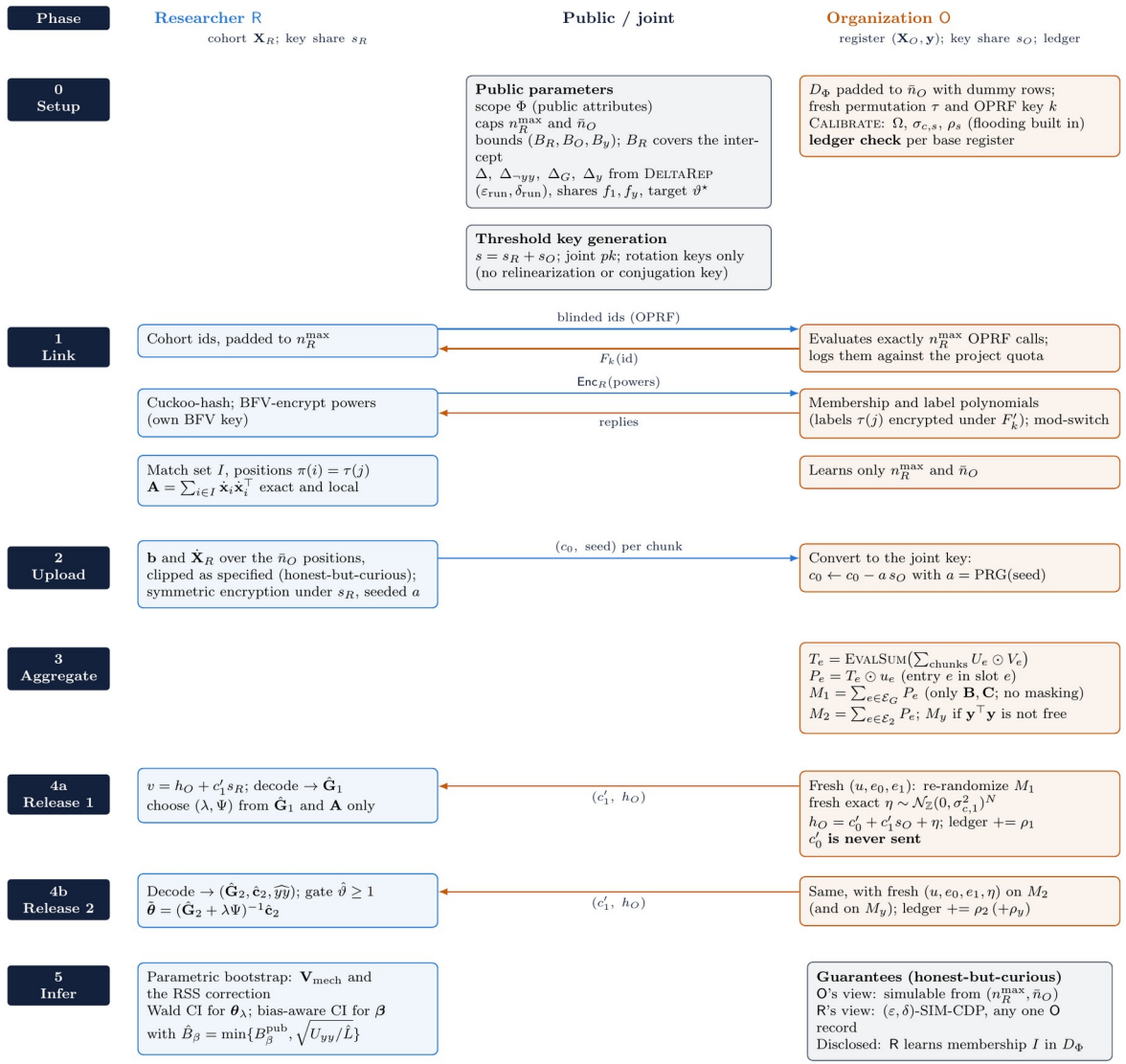
(λ, Ψ) from a small stage-1 release and A only, so it is independent of the stage-2 noise

Valid intervals

Parametric bootstrap; Wald interval for the ridge target θ_λ, bias-aware interval for β

Capped, padded linkage

OPRF-based labeled PSI; R's cohort padded to n_R^{max}, O's register to $\bar{n}_O$



What the proofs establish

Proofs appendix: Part A, cryptography and DP (25 pp.); Part B, sensitivity and statistics (20 pp.)

Part A: cryptography and DP	Status
Encoding sensitivity $\sqrt{(2/N)}$	**Proven**
Smudging lemma, explicit constants	**Proven**
Worst-case error bound $B_{tot} \approx 2^{25.3}$; OpenFHE measures 6–8 bits below it	**Proven**
zCDP of the ideal mechanism; $\rho_{run} = 0.01496$ at $\varepsilon = 1$, $\delta = 10^{-9}$	**Proven**
Output privacy (SIM-CDP) by hybrids, under RLWE and circular security	**Proven**
Input and cohort privacy; correctness without wraparound	**Proven**

Part B: sensitivity and statistics	Status
Replace-one sensitivity, exact for $d_O \geq 2$	**Proven**
$y^T y$ free if $B_y^2 \leq 2(B_R^2 + B_O^2)$, costly if $> 3(B_R^2 + B_O^2)$	**Proven**
Stage-1 ridge choice independent of the stage-2 noise	**Proven**
Operator-norm radius r_1 (one-sided; ~1.55× conservative)	**Proven**
Coverage: Wald remainder exact; plug-in variance; simulated 0.944–0.962	**First order**
Bootstrap RSS correction; removes 87–100% of the bias	**First order**

Still open: proofs for the improved circuits; composition across adaptive runs; a constant-time exact sampler; modulus-chain sizes; the exact free-set boundary for $y^T y$; heteroskedasticity when clipping binds; 16 citation details to check.

What the proofs changed

The CKKS scale was undersized, and several statistical steps need tightening

61.5 bits

CKKS scale needed to flood the worst-case error for the circuit as specified. The v3.1 spec assumed 52.5, which floods only the fresh-encryption error, at its mean.

Circuit	B_{tot}	$\log_2 \Omega$
As specified	$2^{25.3}$	61.5
EvalSum before the first rescale, stochastic rounding, one final rescale	$2^{21.1}$	57.2
Same, with R encoding at scale 2^{60}	$2^{17.6}$	53.7

Corrections to the specification

- Centre the mechanism at the noiseless circuit output; centring at the encoded input would need 67–71 bits
- Calibrate from the data-independent worst-case bound, never from OpenFHE's noise estimate
- The δ slack enters a single (ε, δ) statement with a factor $(1 + e^{\varepsilon})$
- Define β as the coefficient on the clipped data; clipping breaks the homoskedastic variance
- Truncate the bootstrap (its moments do not exist otherwise); spend α_U once in U_{yy}
- The private bound on $\|\beta\|$ is usually infinite while the ridge is on: the β interval then needs a public bound

Takeaways and next steps

1. v3.1 closes the critical gaps for honest-but-curious parties, with proofs; its CKKS scale must rise to about $2^{61.5}$ ($2^{53.7}$ with circuit changes)
2. Within 2× of non-private OLS at n ≈ 3k–100k (simulation); within 10% at 1.4k–11k records for ε = 1–2 (real data)
3. Published AdaSSP is more private than our release and still wins from ε ≈ 0.5; we need about 1.5–2× its records
4. The bias correction is a small effect, now optional; ridge shrinkage dominates when σ is large

Next

- Implement v3.1 in OpenFHE with the improved circuit, and measure its cost
- Rerun the evaluation under v3.1: two-stage ridge, intervals, split bounds
- Fairer AdaSSP comparison: equal μ, a Gaussian-DP budget split, preprocessing that VFL can do
- Close the open proof items; fold the corrections into a v3.2 specification
- Membership-private variant for sensitive registers